

Лабораторная работа № 5

Эмуляция и измерение потерь пакетов в глобальных сетях

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Информация

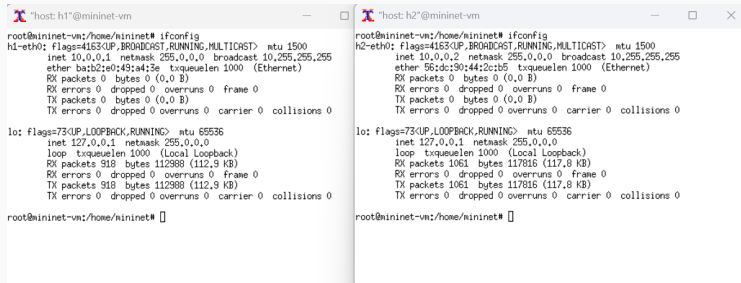
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Основной целью работы является получение навыков проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных. Эти параметры влияют на производительность протоколов и сетей.

1. Задайте простейшую топологию, состоящую из двух хостов и коммутатора с назначенной по умолчанию mininet сетью 10.0.0.0/8.
2. Проведите интерактивные эксперименты по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.
3. Реализуйте воспроизводимый эксперимент по добавлению правила отбрасывания пакетов в эмулируемой глобальной сети. На экран выведите сводную информацию о потерянных пакетах.
4. Самостоятельно реализуйте воспроизводимые эксперименты по исследованию параметров сети, связанных с потерей, изменением порядка и повреждением пакетов при передаче данных. На экран выведите сводную информацию о потерянных пакетах.

Выполнение лабораторной работы



The image shows two terminal windows side-by-side, both titled "host: h1" and "host: h2" respectively, indicating they are running on a Mininet VM. Both windows show the output of the 'ifconfig' command for the 'root@mininet-vn:/home/mininet#' user. The left window (h1) shows the configuration for interface 'h1-eth0' with IP 10.0.0.1 and 'lo' with IP 127.0.0.1. The right window (h2) shows the configuration for interface 'h2-eth0' with IP 10.0.0.2 and 'lo' with IP 127.0.0.1. Both interfaces are in a 'UP' state and have a netmask of 255.0.0.0. The 'lo' interfaces have a netmask of 255.0.0.0 and are in a 'UP' state. The 'h1-eth0' interface has a netmask of 255.0.0.0 and is in a 'UP' state. The 'h2-eth0' interface has a netmask of 255.0.0.0 and is in a 'UP' state. Both interfaces have a txqueuelen of 1000 and show 0 errors, 0 dropped, 0 overruns, 0 frame, 0 TX packets, 0 bytes, 0 TX errors, 0 dropped, 0 overruns, 0 carrier, and 0 collisions.

```
root@mininet-vn:/home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether ba:b2:e0:49:a4:3e txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 918 bytes 112988 (112.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 918 bytes 112988 (112.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vn:/home/mininet#
```

```
root@mininet-vn:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 56:dc:90:44:2c:b5 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1061 bytes 117816 (117.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1061 bytes 117816 (117.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vn:/home/mininet#
```

Figure 1: Информацию о сетевых интерфейсах и IP-адресах хостов

Выполнение лабораторной работы

```
"host: h1"@mininet-vm
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 918 bytes 112988 (112.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 918 bytes 112988 (112.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
34 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.507 ms
34 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.113 ms
34 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.034 ms
34 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.045 ms
34 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.039 ms
34 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.094 ms

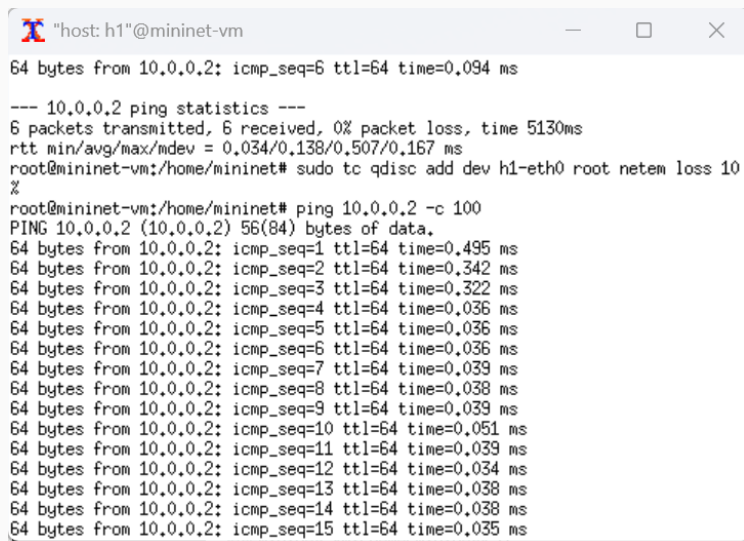
--- 10.0.0.2 ping statistics ---
3 packets transmitted, 6 received, 0% packet loss, time 5130ms
rtt min/avg/max/mdev = 0.034/0.138/0.507/0.167 ms
root@mininet-vm:/home/mininet#

"host: h2"@mininet-vm
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1061 bytes 117816 (117.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1061 bytes 117816 (117.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping -c 6
ping: usage error: Destination address required
root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 6
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=1.18 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.437 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.033 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.033 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.043 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.042 ms

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5105ms
rtt min/avg/max/mdev = 0.033/0.294/1.177/0.420 ms
root@mininet-vm:/home/mininet#
```

Figure 2: Проверка соединения между хостами

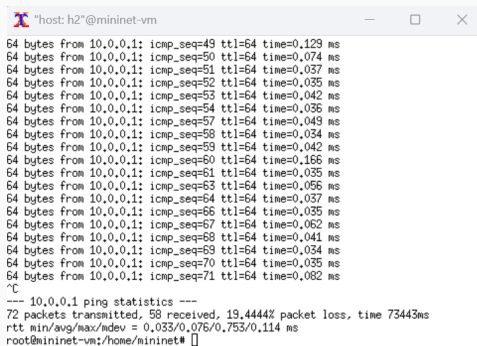


```
"host: h1"@mininet-vm
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.094 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5130ms
rtt min/avg/max/mdev = 0.034/0.138/0.507/0.167 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 10
%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 100
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.495 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.342 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.322 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.036 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.036 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.036 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.039 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.039 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.039 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.034 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.035 ms
```

Figure 3: Добавление потери пакетов

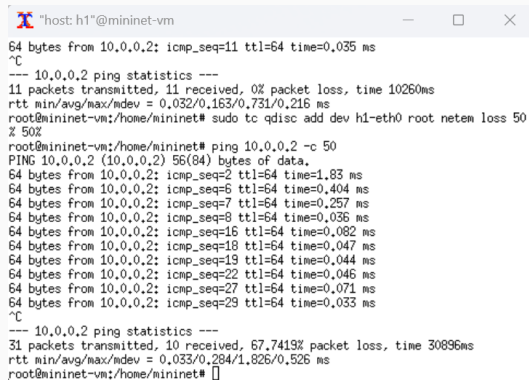
Выполнение лабораторной работы



```
root@mininet-vm:~# ping -c 58 10.0.0.1
64 bytes from 10.0.0.1: icmp_seq=49 ttl=64 time=0.129 ms
64 bytes from 10.0.0.1: icmp_seq=50 ttl=64 time=0.074 ms
64 bytes from 10.0.0.1: icmp_seq=51 ttl=64 time=0.037 ms
64 bytes from 10.0.0.1: icmp_seq=52 ttl=64 time=0.035 ms
64 bytes from 10.0.0.1: icmp_seq=53 ttl=64 time=0.042 ms
64 bytes from 10.0.0.1: icmp_seq=54 ttl=64 time=0.036 ms
64 bytes from 10.0.0.1: icmp_seq=57 ttl=64 time=0.049 ms
64 bytes from 10.0.0.1: icmp_seq=58 ttl=64 time=0.034 ms
64 bytes from 10.0.0.1: icmp_seq=59 ttl=64 time=0.042 ms
64 bytes from 10.0.0.1: icmp_seq=60 ttl=64 time=0.166 ms
64 bytes from 10.0.0.1: icmp_seq=61 ttl=64 time=0.035 ms
64 bytes from 10.0.0.1: icmp_seq=63 ttl=64 time=0.056 ms
64 bytes from 10.0.0.1: icmp_seq=64 ttl=64 time=0.037 ms
64 bytes from 10.0.0.1: icmp_seq=66 ttl=64 time=0.035 ms
64 bytes from 10.0.0.1: icmp_seq=67 ttl=64 time=0.062 ms
64 bytes from 10.0.0.1: icmp_seq=68 ttl=64 time=0.041 ms
64 bytes from 10.0.0.1: icmp_seq=69 ttl=64 time=0.034 ms
64 bytes from 10.0.0.1: icmp_seq=70 ttl=64 time=0.035 ms
64 bytes from 10.0.0.1: icmp_seq=71 ttl=64 time=0.082 ms
^C
--- 10.0.0.1 ping statistics ---
72 packets transmitted, 58 received, 19.4444% packet loss, time 73443ms
rtt min/avg/max/mdev = 0.033/0.076/0.753/0.114 ms
root@mininet-vm:~#
```

Figure 4: Добавление потери пакетов

Выполнение лабораторной работы



```
host: h1" @mininet-vm
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.035 ms
^C
--- 10.0.0.2 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10260ms
rtt min/avg/max/mdev = 0.032/0.163/0.731/0.216 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 50
% 50%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 50
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.83 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.404 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.257 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.036 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.082 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=22 ttl=64 time=0.046 ms
64 bytes from 10.0.0.2: icmp_seq=27 ttl=64 time=0.071 ms
64 bytes from 10.0.0.2: icmp_seq=29 ttl=64 time=0.033 ms
^C
--- 10.0.0.2 ping statistics ---
31 packets transmitted, 10 received, 67.7419% packet loss, time 30896ms
rtt min/avg/max/mdev = 0.033/0.284/1.826/0.526 ms
root@mininet-vm:/home/mininet#
```

Figure 5: Добавление значения корреляции для потери пакетов

Выполнение лабораторной работы

```
"host: h1"@mininet-vm
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem corrupt 0.01%
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 49492 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr    Cwnd
[ 7] 0.00-1.00 sec  5.37 GBytes 46.2 Gbits/sec  10    530 KBytes
[ 7] 1.00-2.00 sec  5.40 GBytes 46.4 Gbits/sec  10    619 KBytes
[ 7] 2.00-3.00 sec  5.09 GBytes 43.8 Gbits/sec   9    486 KBytes
[ 7] 3.00-4.00 sec  5.48 GBytes 47.1 Gbits/sec  19    441 KBytes
[ 7] 4.00-5.00 sec  5.42 GBytes 46.5 Gbits/sec  14    554 KBytes
[ 7] 5.00-6.00 sec  5.04 GBytes 43.3 Gbits/sec  13    390 KBytes
[ 7] 6.00-7.00 sec  5.40 GBytes 46.4 Gbits/sec  11    505 KBytes
[ 7] 7.00-8.00 sec  5.14 GBytes 44.2 Gbits/sec   9    411 KBytes
[ 7] 8.00-9.00 sec  5.34 GBytes 45.8 Gbits/sec  14    518 KBytes
[ 7] 9.00-10.00 sec 4.83 GBytes 41.5 Gbits/sec  13    725 KBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr
[ 7] 0.00-10.00 sec 52.5 GBytes 45.1 Gbits/sec  122
[ 7] 0.00-10.00 sec 52.5 GBytes 45.1 Gbits/sec
sender
receiver

"host: h2"@mininet-vm
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h2-eth0 root netem
root@mininet-vm:/home/mininet# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
Server listening on 5201
-----
Accepted connection from 10.0.0.1, port 49498
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 49492
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-1.00 sec  5.37 GBytes 46.2 Gbits/sec
[ 7] 1.00-2.00 sec  5.40 GBytes 46.4 Gbits/sec
[ 7] 2.00-3.00 sec  5.09 GBytes 43.8 Gbits/sec
[ 7] 3.00-4.00 sec  5.48 GBytes 47.1 Gbits/sec
[ 7] 4.00-5.00 sec  5.42 GBytes 46.5 Gbits/sec
[ 7] 5.00-6.00 sec  5.04 GBytes 43.3 Gbits/sec
[ 7] 6.00-7.00 sec  5.40 GBytes 46.4 Gbits/sec
[ 7] 7.00-8.00 sec  5.14 GBytes 44.2 Gbits/sec
[ 7] 8.00-9.00 sec  5.34 GBytes 45.8 Gbits/sec
[ 7] 9.00-10.00 sec 4.83 GBytes 41.5 Gbits/sec
[ 7] 10.00-10.00 sec 257 KBytes 6.04 Gbits/sec
-----
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-10.00 sec 52.5 GBytes 45.1 Gbits/sec
receiver
```

Figure 6: Добавление повреждения пакетов

Выполнение лабораторной работы

```
"host: h1"@mininet-vm
7] 6.00-7.00 sec 5.40 GBytes 46.4 Gbits/sec 11 505 KBytes
7] 7.00-8.00 sec 5.14 GBytes 44.2 Gbits/sec 9 411 KBytes
7] 8.00-9.00 sec 5.34 GBytes 45.8 Gbits/sec 14 518 KBytes
7] 9.00-10.00 sec 4.83 GBytes 41.5 Gbits/sec 13 725 KBytes
-----
ID] Interval      Transfer      Bitrate      Retr
7] 0.00-10.00 sec 52.5 GBytes 45.1 Gbits/sec 122
7] 0.00-10.00 sec 52.5 GBytes 45.1 Gbits/sec
sender
receiver

iperf Done.
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem corrupt
0.01%
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
7] local 10.0.0.1 port 40500 connected to 10.0.0.2 port 5201
ID] Interval      Transfer      Bitrate      Retr      Cwnd
7] 0.00-1.00 sec 5.58 GBytes 47.9 Gbits/sec 65 448 KBytes
7] 1.00-2.00 sec 5.32 GBytes 45.7 Gbits/sec 19 772 KBytes
7] 2.00-3.00 sec 5.23 GBytes 44.9 Gbits/sec 13 369 KBytes
7] 3.00-4.00 sec 5.46 GBytes 46.9 Gbits/sec 13 303 KBytes
7] 4.00-5.00 sec 5.18 GBytes 44.5 Gbits/sec 15 266 KBytes
7] 5.00-6.00 sec 5.49 GBytes 47.2 Gbits/sec 19 291 KBytes
7] 6.00-7.00 sec 5.51 GBytes 47.3 Gbits/sec 16 563 KBytes
7] 7.00-8.00 sec 5.02 GBytes 43.1 Gbits/sec 13 509 KBytes
7] 8.00-9.00 sec 5.35 GBytes 46.0 Gbits/sec 44 407 KBytes
7] 9.00-10.00 sec 5.17 GBytes 44.4 Gbits/sec 13 413 KBytes
-----
ID] Interval      Transfer      Bitrate      Retr
7] 0.00-10.00 sec 53.3 GBytes 45.8 Gbits/sec 230
7] 0.00-10.00 sec 53.3 GBytes 45.8 Gbits/sec
sender
receiver

iperf Done.
root@mininet-vm:/home/mininet#
```

Figure 7: Добавление переупорядочивания пакетов

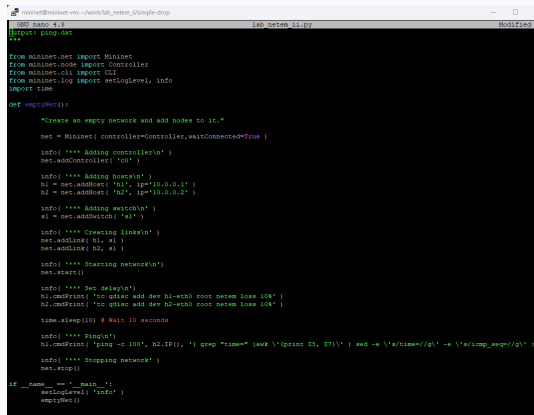
Выполнение лабораторной работы

```
"host: h1"@mininet-vm
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem duplica
te 50%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 20
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data,
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.54 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.51 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.74 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.219 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.038 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.145 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.147 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.065 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.066 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.042 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.043 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.034 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.154 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.141 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.142 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.090 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.092 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.037 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.031 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.031 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.041 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.042 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.038 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.062 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, +10 duplicates, 0% packet loss, time 19438m
s
rtt min/avg/max/mdev = 0.031/0.259/2.538/0.575 ms
root@mininet-vm:/home/mininet#
```

Figure 8: Добавление дублирования пакетов

Выполнение лабораторной работы



```
mininet@mininet-vm: ~/work/lab_netem_II/simple-drop
GNU nano 4.8 lab_netem_ii.py Modified
Output: ping.dat
***
from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
import time

def emptyNet():
    "Create an empty network and add nodes to it."
    net = Mininet( controller=Controller, waitConnected=True )
    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s1 = net.addSwitch( 's1' )

    info( '*** Creating links\n' )
    net.addLink( h1, s1 )
    net.addLink( h2, s1 )

    info( '*** Starting network\n' )
    net.start()

    info( '*** Set delay\n' )
    h1.cmdPrint( 'tc qdisc add dev h1-eth0 root netem loss 10%' )
    h2.cmdPrint( 'tc qdisc add dev h2-eth0 root netem loss 10%' )

    time.sleep(10) # Wait 10 seconds

    info( '*** Ping\n' )
    h1.cmdPrint( 'ping -c 100', h2.IP(), '| grep "time=" | awk '{print $5, $7}' | sed -e \"s/time=//g\" -e \"s/icmp_seq=//g\" )

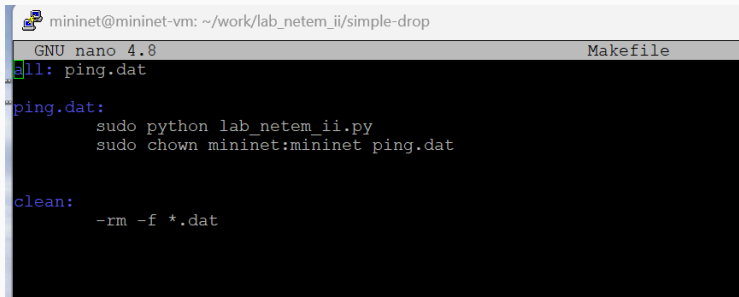
    info( '*** Stopping network' )
    net.stop()

if __name__ == '__main__':
    setLogLevel( 'info' )
    emptyNet()
```

Figure 9: Создание скрипта для эксперимента lab_netem_ii.py

```
hl.cmdPrint('ping -c 100 10.0.0.2 | grep "packet loss:" | awk \'{print $6, $7, $8}\' > ping.dat')
```

Figure 10: Редактирование скрипта



The image shows a terminal window with a nano editor interface. The title bar at the top reads "mininet@mininet-vm: ~/work/lab_netem_ii/simple-drop". The editor shows a file named "Makefile". The content of the Makefile is as follows:

```
GNU nano 4.8 Makefile
all: ping.dat

ping.dat:
    sudo python lab_netem_ii.py
    sudo chown mininet:mininet ping.dat

clean:
    -rm -f *.dat
```

Figure 11: Makefile

Выполнение лабораторной работы

```
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ make
sudo python lab_netem_ii.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem loss 10%',)
*** h2 : ('tc qdisc add dev h2-eth0 root netem loss 10%',)
*** Ping
*** h1 : ('ping -c 100 10.0.0.2 | grep "packet loss:" | awk \'{print $6, $7, $8}\'' > ping.dat',)
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$
```

Figure 12: Проведение эксперимента

В результате выполнения данной лабораторной работы я получил навыки проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.

1. Mininet [Электронный ресурс]. Mininet Project Contributors. URL: <http://mininet.org/> (дата обращения: 17.11.2024).